Technical Service Bulletin

Technical Service Bulletin: TSB210241	**Released Date:** 20-Jul-2026
Elevated Copper Level in Engine Lubricating Oil	

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- K19 (All Versions)
- QSK19 (All Versions)
- QSK23 (All Versions)
- QST30 (All Versions)
- K38 (All Versions)
- QSK38 (All Versions)
- QSK45 (All Versions)
- QSK50 (All Versions)
- QSK60 (All Versions)
- QSK78 (All Versions)
- QSK95 (All Versions)
- V28 (All Versions)
- V903 (All Versions)

Issue Summary

Symptom:

- Elevated copper levels observed in engine lubricating oil analysis reports

Verification

Verify copper levels (without change of trends in iron and lead) are above historical statistical mean for given engine lubricating oil in analysis report.

Resolution

- Check for trend of other wear metals in engine lubricating oil, such as iron and lead.
- If other wear metals are also trending upward, reference Service Bulletin, Oil Analysis Techniques for High Horsepower Diesel Engines, Bulletin 4022060 (/qs3/pubsys2/xml/en/bulletin/4022060.html).

- If copper is the **only** wear metal trending high and levels are above historical statistical mean but below level of 180 ppm, continue normal engine lubricating oil sampling schedule.
- If copper levels are above 180 ppm engine lubricating oil should be drained and replaced.

Description of Change

- Copper can leach out of lubricating oil coolers and cause copper content to rise in engine lubricating oil. This typically is **not** cause for concern unless other wear metals are elevated, which could indicate a component issue. Over time copper levels return to normal as the copper will have fully leached from the lubricating oil cooler. This time can vary depending on engine run time, duty cycle, and engine lubricating oil formulation.
- Engage local technical support channel if the issue persists or there are any concerns.

Document History

Date	Details
2021-10-29	Module Created
2023-3-22	Updated Product Affected section.
2026-7-16	Updated Product Affected section.